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“Free” Hot Water from a Wood Heater

Though, in my nearly 40 years of construction practice, I’ve never either designed or built an off-grid home, I’ve known a number of people who lived that more self-sufficient lifestyle, and occasionally helped with their projects.

In 2009, for instance, I helped the Arch-Druid of Vermont lay down the final (of three) layers of an earthen floor inside his circular cord-wood masonry house with living roof complete with beehives (to keep them safe from bears) just outside of Montpelier Vermont.



Terracotta Topped Earthen Floor – Water Coil on Flue Pipe to Preheat for Electric Tank Behind

And I was first drawn back to Maine in 2015 when I did an on-site consultation for an addition to an owner-built octagonal outward leaning wooden yurt that was the core of a young couple's off-grid homestead.

But my greatest engineering challenge came in the autumn of 2016, when I was staying with a recently-widowed homesteader living in an owner-built rectangular cordwood masonry home in central Maine, that had two solar PV panels to supply 12-volt LED lights and power a chest-type high-efficiency refrigerator.

Her heat was provided by a custom-built masonry wood heater, and her water came from a drilled well pumped by hand with a Maine-made stainless steel Bison pitcher pump. Hot water had to be heated on the propane cookstove and dumped over one's head in the clawfoot tub.



Cordwood Masonry Home in Maine on Wooden Grade Beam Foundation

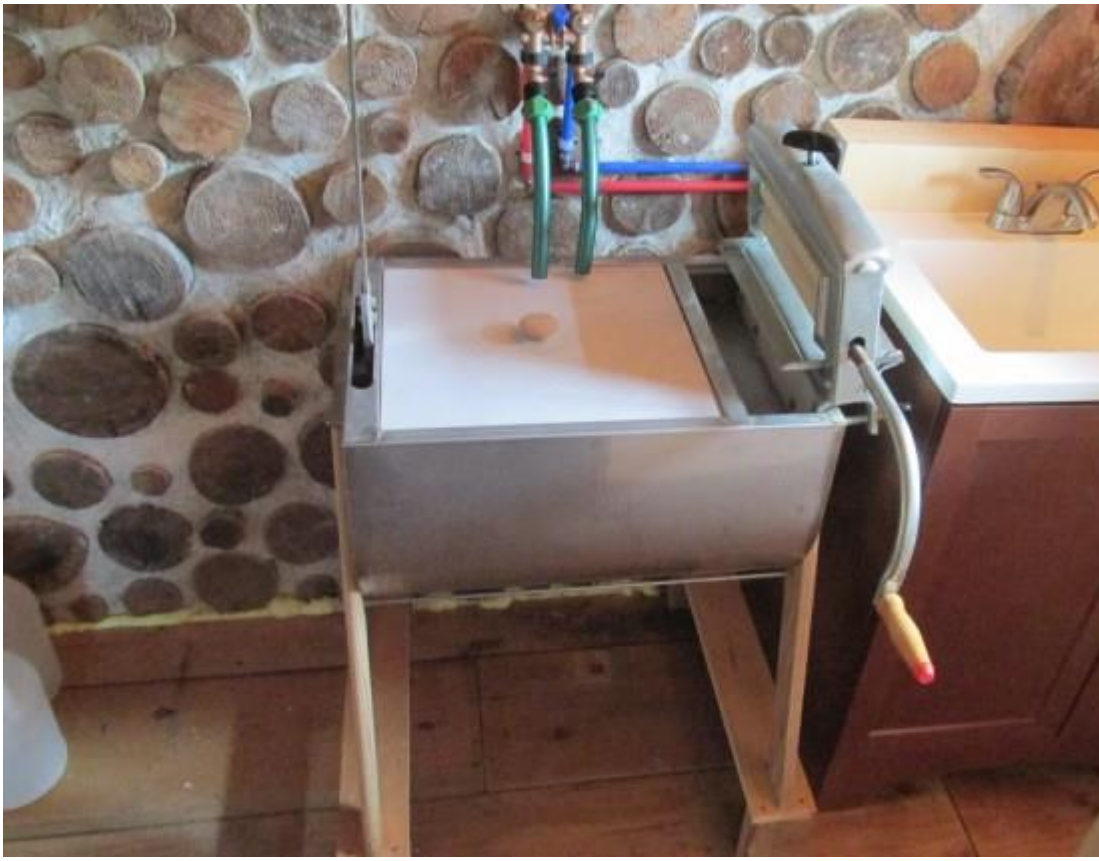
The house did have a simple gray-water system to absorb the effluent from the kitchen sink and bathtub. Perhaps because she was then doing all her farmstead chores alone (or perhaps because she was thinking of eventually selling the place), my friend wanted the luxury of running water and a hot shower.

I finished installing a 12-volt RV water pump, tapped into the Bison pump's intake line. Normally, the well water rests well up in the well casing about 15′ below ground. This pump was rated for just 15′ of head, but we were pleasantly surprised that it worked until a dry summer came

along, but even then could be primed by a couple of swings of the Bison pump handle.

Once we knew that cold water would flow, I installed a 6-gallon propane-fired RV water heater in an insulated box outside the house at the kitchen sink wall. The hexagonal tank was well insulated with molded rigid foam and the additional mineral wool batting allowed the heater to maintain its heat during a cool spring or fall night, with only the pilot light to keep it up to the set-point temperature.

With hot water plumbed into the house system, I then removed the old clawfoot tub and built a shower surround, installed a lavatory sink and plumbed both supply and drain lines for an Amish-built stainless steel hand-operated clothes washing tub.



With new PEX water supply lines and PVC drain and vent lines, the system was almost modern. (A couple of years later, I had to dig up and rebuild the exterior drain line to the gray water leaching bed, as it was installed with insufficient slope and had clogged.)



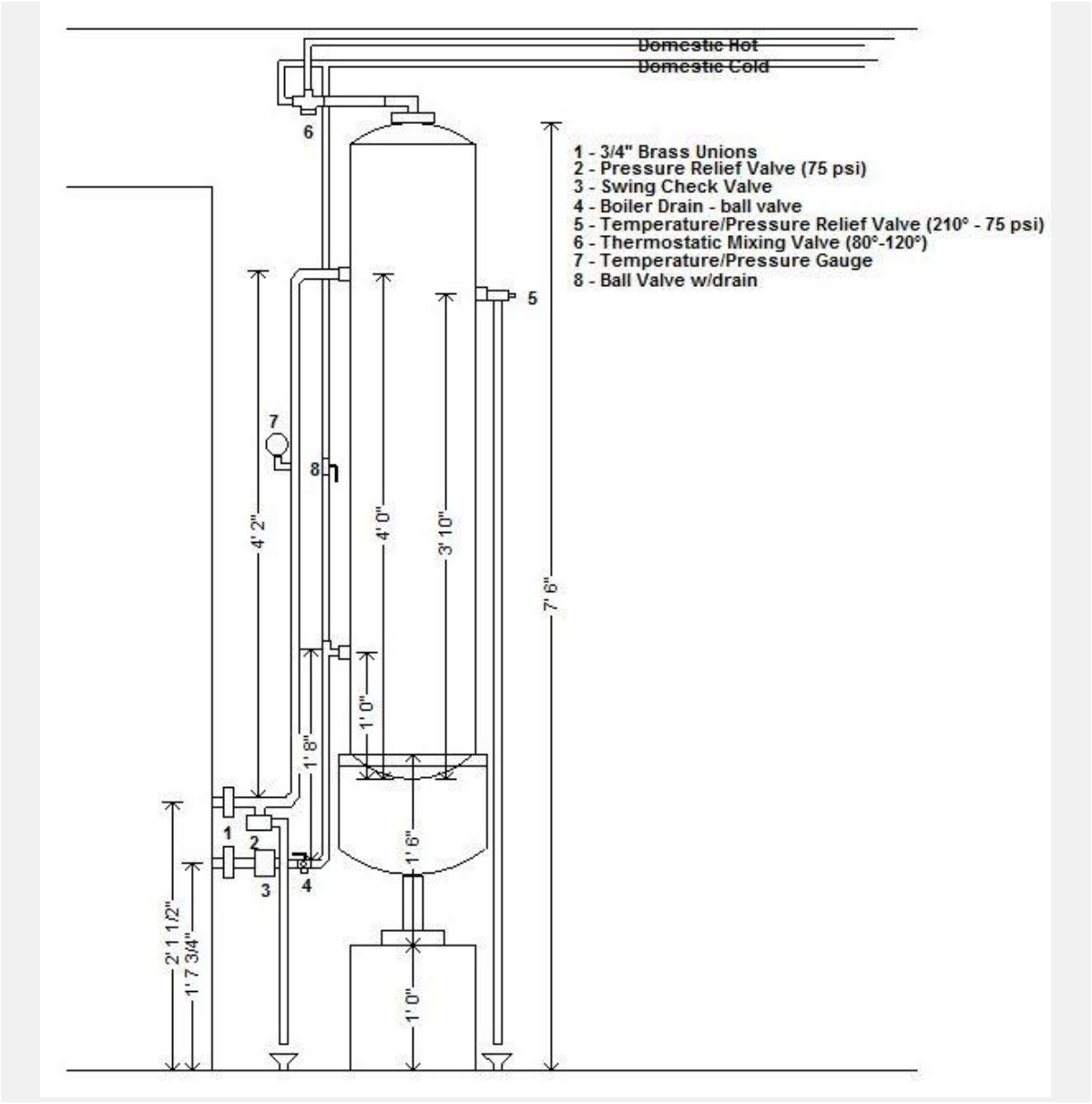
Those improvements took care of summertime hot water. But, with the propane-fired hot water tank outside the house, it had to be shut off and drained before freeze-up.

My friend had been prescient enough to have a stainless steel U-shaped heat-exchange pipe installed in the firebox when the masonry heater was built, and her late husband had scrounged a 30-gallon copper hot water tank which had been pressure-tested for leaks. She then asked me to design and build a thermo-siphon hot water system for winter use when the masonry heater was providing warmth for the house.

As per my habit, I engaged in an internet search for examples and standards for such a wood-fired water heater and was surprised to find that there was very little reliable information on the World Wide Web.

Perhaps it should not have been a surprise, since no plumbing code I'm aware of (and I used to teach plumbing) deals with off-grid thermo-siphon systems. Thus few, if any, licensed plumbers would have any experience with them. The few examples I found on-line were owner-built and of questionable quality.

But I gleaned what I could and combined it with my expertise in Hygro-Thermal Engineering (another class I used to teach) to create a custom-designed unit for this off-grid homestead.



Schematic

Her husband had also left her a classic iron tank pedestal that, placed atop a wooden base, kept the tall copper tank above the wood heater loop and close enough to the ceiling to optimize the stack-effect for passive thermal siphoning.



Generally, a minimum height difference between cold water outlet and hot water inlet of 3′ is sufficient to start and sustain a thermosiphon loop. I managed to get more than 4′ of head and used 1″ copper piping to minimize internal flow resistance.

Since the tank was stamped 80 PSI maximum, I installed both the standard temperature/pressure relief valve on the tank and an additional 75 PSI pressure relief valve on the supply side of the heat exchange loop from the wood heater to make sure than there would be no overpressure problems, which can be explosive when the heat source is a wood fire burning at several hundred degrees.

I installed an easily visible combination temperature and pressure gauge to monitor the system's operation, and used an adjustable thermal mixing valve to temper the hot water to the house plumbing in case the temperature rose to high in the tank during heavy use of the heater (which is fired only one or twice per day but at full open burn).



Complete System with Pop-Off Drains Through Floor & Mixing Valve at Top



Though I used all my decades of hard-earned expertise, because there were really no blueprints for such a system, and because each system must be designed for a particular application, I did not know until we fired it up if it would work as designed.

But, in fact, it exceeded our expectations, providing quick and sufficient hot water all during the heating season without ever reaching excessive temperature or pressure. The only thing remaining to add, because the copper tank is so highly conductive, was a blanket of insulation to retain the water's temperature in between firings of the wood heater (which might be only once in 24 hours).

And, because it was so difficult to find a reliable thermosiphon system to replicate for this off-grid homestead, I wanted to share my design, experience and results with my readers.